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Adaptive Scaffolding with Generative AI Chatbot for Solving Ill-Structured Problems

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Abstract: This study investigates the role of generative AI (GenAI) chatbots in supporting students' ill-structured problem-solving. A GenAI chatbot, *Probot*, was developed to offer adaptive scaffolding through question prompts during problem representation, solution generation and selection, justification, and monitoring and evaluation. Sixty-one undergraduate students in Korea were divided into control and experimental groups. The experimental group solved an ill-structured problem with *Probot* while the control group used a non-adaptive chatbot. Interaction patterns with GenAI chatbots were analyzed using Epistemic Network Analysis. Results showed the experimental group had strong connections with orientation in problem analysis, solution analysis, and solution elaboration, suggesting that *Probot* supported students' analytical and elaborative thinking. The control group showed strong connections between problem analysis, solution analysis, and solution suggestions, indicating a focus on information and idea-seeking. The findings highlight *Probot*'s potential in guiding ill-structured problem-solving and offer design insights for adaptive scaffolding.

Introduction

Problem-solving skills have emerged as a critical competency for students in the 21st century (Partnership for 21st Century Skills, 2015). Among various types of problems, students particularly face difficulties in solving ill-structured problems due to their abstract and complex nature, which requires advanced cognitive skills (Jonassen, 2000). In general, the ill-structured problem-solving process is presented as a series of phases, namely problem representation, generating and selecting solutions, making justifications, and monitoring and evaluating (Ge & Land, 2004). Given that ill-structured problem-solving demands extensive cognitive and metacognitive skills, scaffolding can be valuable for students who encounter difficulties in the ill-structured problem-solving process.

Prior studies on the mechanism to support students in solving ill-structured problems have primarily employed fixed conceptual and metacognitive scaffolding. Such scaffolding is provided to help students who struggle to identify relevant information due to the lack of necessary conceptual knowledge (Kim & Lim, 2019) and to effectively utilize metacognitive skills in solving ill-structured problems (Ge & Land, 2004). For instance, Kim and Lim (2019) provided scaffolding by posting exploratory questions, hints, and visual materials on a bulletin board for solving instructional design problems. Ouyang et al. (2021) offered students voice- and text -based scaffolding through an online platform to solve the practical challenges instructors face in remote online education during COVID-19.

Some studies have explored the use of adaptive scaffolding in ill-structured problem-solving. For instance, Winkler et al. (2021) implemented adaptive scaffolding by providing feedback through a personal AI assistant based on students' solutions to legal problems, while Zheng et al. (2022) offered adaptive scaffolding tailored to students' performance in solving C programming problems. However, since adaptive scaffolding messages are often pre-designed by experts or instructors, such scaffolding is limited in its ability to fully address individual students' needs. Additionally, these scaffolding approaches are typically confined to specific domains, such as instructional design, legal, or programming problems. This highlights the need for further research into adaptive scaffolding mechanisms that dynamically respond to students' specific questions or interactions, rather than relying solely on pre-designed messages or narrowly focused problem domains.

Generative AI (GenAI), which leverages deep learning models to create diverse content such as images and text, offers significant potential for dynamically generating scaffolding customized based on the context of problems and student input (Lim et al., 2023). In this study, we developed a GenAI-integrated chatbot, *Probot*, to provide adaptive conceptual and metacognitive scaffolding to support students in solving ill-structured problems. This research investigates how students interact with *Probot* throughout the problem-solving process. University students who participated in the study were divided into two groups: one with adaptive scaffolding provided by *Probot* and the other without adaptive scaffolding. Specifically, the study addresses the following research questions: (a) What patterns emerge when students interact with *Probot* during the ill-structured problem-solving process? (b) How do these interaction patterns differ between the groups with and without adaptive scaffolding?

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Methods

Research context and participants

This study recruited 61 Korean undergraduate students (25 males and 36 females) through convenience sampling and assigned them to either an experimental group (31 students) or a control group (30 students). Participants first completed a one-hour pre-task a week before the experiment, which involved proposing a policy to address issues related to deepfake technology. The task required students to define the problem, propose a policy, justify their solution, and monitor and evaluate their proposed solution in essay format. Two independent coders graded the outputs using rubric, which assesses the problem and cause analysis, solution generation, solution justification, and solution monitoring and evaluation. Cohen's kappa coefficient was .87. Based on the pre-task scores, participants were evenly distributed between the control and experimental groups. A Levene test (p = .96) confirmed the homogeneity of the two groups.

The main experiment took place in a university computer lab where participants solved an ill-structured problem by proposing a policy to address issues related to teenagers' use of social networking sites (SNS). Students were required to define the problem, justify their proposed solution, and monitor and evaluate their approach, producing their answers in essay format. The control group used *TalkGPT*, a GenAI chatbot without adaptive scaffolding. The experimental group used *Probot*, which provided adaptive scaffolding tailored to students' interactons with *Probot*.

Probot: GenAI chatbot for ill-structured problems

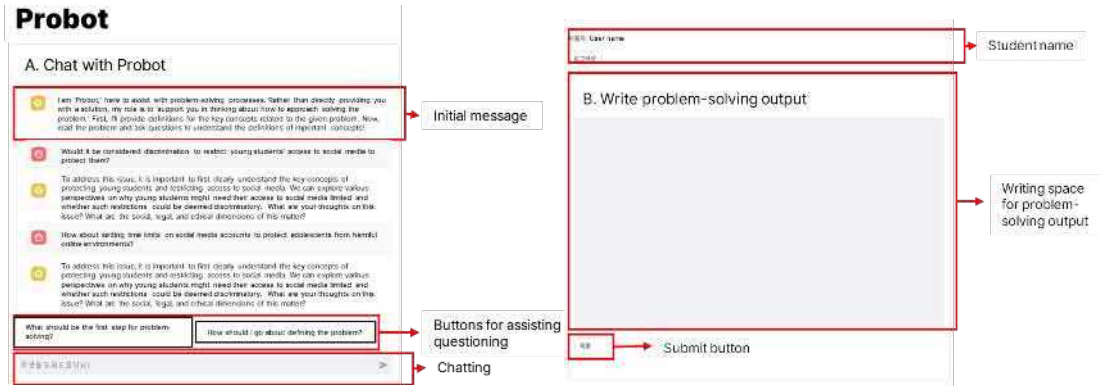
This study developed a GenAI chatbot, *Probot*, to provide conceptual and metacognitive scaffolding in the form of question prompts to support students in solving ill-structured problems. Drawing on prior research, the scaffolding was carefully designed to address the specific needs of each phase of the ill-structured problem-solving process: (a) problem representation, (b) generating and selecting solutions, (c) making justifications, and (d) monitoring and evaluating (Ge & Land, 2004; Kim & Lim, 2019; Ouyang et al., 2021; Winkler et al., 2021).

In the problem representation phase, conceptual scaffolding was designed to help students identify key concepts of the problem, elaborate their thoughts, and understand the relationships between different problem domains. Additionally, metacognitive scaffolding supported students in accessing their prior knowledge about the problem and monitoring and evaluating their progress in understanding and framing the issue. For the generating or selecting solutions phase, conceptual scaffolding was used to encourage students to elaborate their ideas and articulate their opinions about the problem. Metacognitive scaffolding, on the other hand, guided students to refine their thoughts and reflect on the feasibility and relevance of their proposed solutions, promoting deeper engagement with the task. During the making justifications phase, conceptual scaffolding played a crucial role in helping students infer logical connections and justify their proposed solutions. This type of scaffolding encouraged students to develop well-supported arguments and strengthen the validity of their solutions. Finally, in the monitoring and evaluating phase, metacognitive scaffolding supported students in critically assessing their proposed solutions. This involved encouraging them to monitor their decision-making process and evaluate the quality and effectiveness of their solutions.

We developed *Probot* using prompt engineering with a Large Language Model (LLM), specifically gpt-4-turbo-2024-04-09. In designing *Probot*, we applied principles from Bsharat et al. (2024) such as example-driven prompting, which involved assigning a specific task and role to the chatbot. The prompts were developed to provide scaffolding without directly offering answers to engage students in the problem-solving process. For the user interface, we built the chatbot using Python and integrated LangChain, Streamlit, Streamlit Cloud, and Firebase. These tools allowed to connect to the LLM and collect log data from the interactions between students and *Probot*. *Figure 1* shows the interface of *Probot* and sample conversation logs demonstrating how participants used *Probot* to solve ill-structured problems. To investigate differences in student interactions with GenAI chatbots, the control group used *TalkGPT*, an identical chatbot interface without adaptive scaffolding. *TalkGPT* maintains a direct response mechanism similar to ChatGPT to ensure that interactions closely resembled a typical user experience with GenAI chatbots.

Figure 1
Probot Interface and Conversation Example between Probot and Participant

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Data analysis

To analyze interaction patterns through conversational logs, we employed the Functional Category System (FCS) developed by Poole and Holmes (1995). The FCS coding scheme, shown in *Table 1*, includes four main categories of students' ill-structured problem-solving process: problem representation, orientation, solution development, and others. Duplicate coding was permitted for each conversation to ensure a comprehensive analysis. Two independent coders applied the scheme to the conversation logs, achieving a Cohen's kappa coefficient of .82. Any discrepancies between the coders were resolved through discussion until consensus was reached. After coding the data, we conducted an Epistemic Network Analysis (ENA) to visualize interaction patterns and compare the networks between the control and experimental groups (Shaffer et al., 2016). Using the ENA web tool, we set the control and experimental groups as units of analysis, the students' conversation logs with GenAI chatbots as the conversation source, and a stanza window size of 4. To evaluate differences between the two groups, we performed a two-sample *t*-test to identify statistically significant differences along the axis in the ENA.

Table 1

Coding Scheme of the Ill-structured Problem-Solving Process

Category	Sub-category	Code	Code description
Problem representation	Problem Analysis	PA	explain or identify the underlying causes of a problem
	Problem Critique	PC	evaluate the analysis of the problem
Orientation	Orientation	OO	direct or guide the problem-solving process
	Process Reflection	PR	review or evaluate the problem-solving progress
Solution Development	Solution Analysis	SA	address decision-making criteria or outline general guidelines for solutions
	Solution Suggestion	SS	suggest alternative solutions
	Solution Elaboration	SE	expand on or add detail to a previously mentioned alternative. offer additional ideas or information about alternative solutions
	Solution Evaluation	SV	evaluate alternatives and provide either explicit or implicit reasons for evaluations
	Solution Confirmation	SC	present the final decision or request confirmation of the decision
Others	Non-Task	NT	unrelated to the decision task
	Simple Agreement	SG	simply agree with a statement in the problem-solving process
	Simple Disagreement	SD	simply disagree with a statement in the problem-solving process

Results

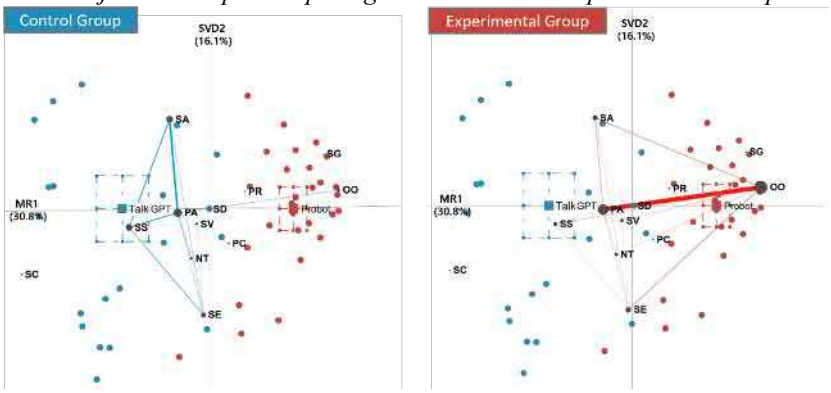
We examined the interaction between *Probot* and participants in ill-structured problem-solving through ENA to identify how chatbots support students' ill-structured problem-solving. The control and experimental groups' ENA projection graphs are presented in **Figure 2**. The model showed that MR1 (means rotation) dimensions accounted for 30.8% while SVD2 (singular value decomposition) dimensions accounted for 16.1% of the variance in the data. Along the x-axis (MR1), the two-sample *t*-test result showed a statistically significant difference ($t(44.05)=11.89$, $p=0.01$, Cohen's $d=3.07$) between the control (mean=-0.35, $SD=0.28$) and experimental

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(mean=0.34, *SD*=0.15) groups. In the y-axis (SVD2), no statistically significant differences were observed. As illustrated in **Figure 2**., the experimental group showed strong connections between Orientation (OO) and problem analysis (PA) with a connection weight of 0.72. In sequence, orientation (OO) and solution elaboration (SE), orientation (OO) and solution analysis (SA) showed strong connections (connection weight=0.20 and 0.19). On the other hand, for the control group, strong connections between problem analysis (PA) and solution analysis (SA) (connection weight =0.29), problem analysis (PA) and solution suggestion (SS) (connection weight=0.20), and solution analysis (SA) and solution suggestion (SS) (connection weight=0.20) were observed.

Figure 2

ENA Projection Graph Comparing the Control and Experimental Groups



Discussion and conclusion

This study used ENA to explore how *Probot*, a GenAI chatbot with adaptive scaffolding, can support undergraduate students' ill-structured problem-solving. The main findings include the following. First, the experimental group using *Probot* demonstrated strong connections between orientation and problem analysis, indicating that students spent substantial time in problem representation. *Probot* also provided significant direction and guidance during problem analysis, as well as during solution analysis and solution elaboration. This pattern suggests that *Probot* played a crucial role in guiding students' analytical and elaborative thinking processes. Second, the control group showed strong connections between problem analysis, solution analysis, and solution suggestion, indicating that students mainly sought help in problem identification and solution generation. This result implies that students tend to seek information and solution ideas in their ill-structured problem-solving simultaneously. This pattern is similar to that observed by Krupp et al. (2024) and Yan (2023), in which students request the output generated by ChatGPT and use them for assigned tasks. Additionally, the absence of strong connections with orientation implies that *TalkGPT* lacks guiding ill-structured problem-solving processes. This may be due to the tendency of GenAI chatbots like ChatGPT to adopt a prescriptive approach, producing outputs that match students' requests. (Contel & Cusi, 2024). Third, the main difference between the two groups was the presence of orientation in the experimental group. It demonstrates that students who interacted with *Probot* received scaffolding messages supporting more time in analyzing the problem, which can enhance students' comprehension of the given problems. This interaction pattern supports Fui-Hoon Nah et al.'s (2023) claim that GenAI could facilitate learners' brainstorming in problem-solving. As the context, level of abstraction, and complexity of ill-structured problem-solving are high (Jonassen, 2000), the problem representation phase becomes crucial. For future chatbot design, segmented scaffolding in problem representation to support a more in-depth analysis phase such as identifying key concepts, stakeholders, and constraints related to the ill-structured problem is recommended.

This study has some limitations that should be acknowledged. First, while the groups were divided into experimental and control groups for ENA, further analysis based on students' problem-solving performance levels could provide more insights into interaction patterns adaptive to varying skill levels. Second, the scaffolding provided by *Probot* was limited to cognitive scaffolding for conceptual and metacognitive understanding. Future research could expand on this by incorporating affective scaffolding to address students' socio-emotional needs during the ill-structured problem-solving process. Despite these limitations, this study makes a valuable contribution to understanding adaptive scaffolding in problem-solving. Developing *Probot* using question prompts demonstrates the potential of GenAI in dynamically scaffolding the ill-structured problem-solving process.

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